

SPECIALIZED_PLATFORMS_QUICKSTART

HelixCode Specialized Platforms - Quick Start Guide

Overview

This guide helps you quickly get started with HelixCode on Aurora OS (Russian platform) and Harmony OS (Chinese platform). Both platforms offer specialized features optimized for their respective ecosystems.

Quick Platform Comparison

Feature	Aurora OS	Harmony OS
Target Market	Russian Federation	China / Global
Focus	Security & System Integration	Distributed Computing & AI
Key Features	Enhanced security, system monitoring	NPU acceleration, cross-device sync
Hardware Support	Standard x86/ARM	Kirin chipsets, NPU/GPU
Best For	Secure environments, government	Multi-device workflows, AI workloads

Aurora OS - Quick Start

5-Minute Setup

Step 1: Build

```
cd HelixCode
make aurora-os
```

Output: bin/aurora-os (approximately 56 MB)

Step 2: Deploy (Automated)

```
sudo ./scripts/deploy-aurora-os.sh
```

This script automatically: - Installs binary to /opt/helixcode/ - Creates configuration files - Sets up systemd service - Configures permissions

Step 3: Configure Database

Edit /etc/helixcode/aurora.env:

```
HELIX_DATABASE_PASSWORD=your_secure_password
HELIX_AUTH_JWT_SECRET=your_jwt_secret_key
```

Create database:

```
sudo -u postgres createdb helixcode
sudo -u postgres createuser helixcode
sudo -u postgres psql -c "ALTER USER helixcode WITH PASSWORD
    'your_secure_password';"
```

Step 4: Start Service

```
sudo systemctl start helixcode-aurora
sudo systemctl enable helixcode-aurora
```

Step 5: Verify

```
# Check status
sudo systemctl status helixcode-aurora
```

```
# View logs
sudo journalctl -u helixcode-aurora -f
```

```
# Test health endpoint
curl http://localhost:8080/health
```

Done! Access the UI at <http://localhost:8080>

Quick Configuration

Minimal /etc/helixcode/aurora-config.yaml:

```
server:
  address: "0.0.0.0"
  port: 8080
```

```
database:
  host: "localhost"
  port: 5432
  user: "helixcode"
  dbname: "helixcode"

aurora:
  enable_security_features: true
  enable_system_monitoring: true
  security_level: "enhanced"

workers:
  max_concurrent_tasks: 10

logging:
  level: "info"
  file: "/var/log/helixcode/aurora-os.log"
```

Common Aurora OS Commands

```
# Start/stop service
sudo systemctl start helixcode-aurora
sudo systemctl stop helixcode-aurora

# View logs
sudo journalctl -u helixcode-aurora -f

# Check configuration
helixcode-aurora --config /etc/helixcode/aurora-config.yaml --
  validate

# System diagnostics
helixcode-aurora diagnostics

# Security status
helixcode-aurora security status
```

Harmony OS - Quick Start

5-Minute Setup

Step 1: Build

```
cd HelixCode
make harmony-os
```

Output: bin/harmony-os (approximately 56 MB)

Step 2: Deploy (Automated)

```
sudo ./scripts/deploy-harmony-os.sh
```

This script automatically: - Installs binary to /opt/helixcode/ - Creates configuration files - Sets up systemd service or init script - Configures directories and permissions

Step 3: Configure Database & Redis

Edit /etc/helixcode/harmony.env:

```
HELIX_DATABASE_PASSWORD=your_secure_password
HELIX_REDIS_PASSWORD=your_redis_password
HELIX_AUTH_JWT_SECRET=your_jwt_secret_key
```

Create database:

```
sudo -u postgres createdb helixcode
sudo -u postgres createuser helixcode
sudo -u postgres psql -c "ALTER USER helixcode WITH PASSWORD
    'your_secure_password';"
```

Install Redis (optional but recommended):

```
sudo apt install redis-server
sudo systemctl enable redis-server
sudo systemctl start redis-server
```

Step 4: Start Service

```
# If systemd is available
sudo systemctl start helixcode-harmony
sudo systemctl enable helixcode-harmony
```

```
# Or using init script
sudo /etc/init.d/helixcode-harmony start
```

Step 5: Verify

```
# Check status
sudo systemctl status helixcode-harmony
```

```
# View logs
sudo journalctl -u helixcode-harmony -f
```

```
# Test health endpoint
curl http://localhost:8080/health
```

```
# Check distributed engine
helixcode-harmony distributed status
```

Done! Access the UI at <http://localhost:8080>

Quick Configuration

Minimal `/etc/helixcode/harmony-config.yaml`:

```
server:
  address: "0.0.0.0"
  port: 8080

database:
  host: "localhost"
  port: 5432
  user: "helixcode"
  dbname: "helixcode"

redis:
  enabled: true
  host: "localhost"
  port: 6379

harmony:
  enable_distributed_computing: true
  enable_cross_device_sync: true
  enable_ai_acceleration: true
  enable_resource_optimization: true

workers:
```

```
max_concurrent_tasks: 20
distributed_mode: true
```

```
logging:
  level: "info"
  file: "/var/log/helixcode/harmony-os.log"
```

Common Harmony OS Commands

```
# Start/stop service
sudo systemctl start helixcode-harmony
sudo systemctl stop helixcode-harmony

# View logs
sudo journalctl -u helixcode-harmony -f

# Distributed computing
helixcode-harmony distributed status
helixcode-harmony distributed workers

# Device info
helixcode-harmony device list
helixcode-harmony device npu-status

# Synchronization
helixcode-harmony sync status
helixcode-harmony sync now
```

Both Platforms - Combined Deployment

Deploy Both at Once

```
# Build both platforms
make aurora-harmony

# Deploy both (interactive)
sudo ./scripts/deploy-specialized-platforms.sh

# Or specify both explicitly
sudo ./scripts/deploy-specialized-platforms.sh --platform both --
      build
```

Managing Both Services

```
# Start both
sudo systemctl start helixcode-aurora helixcode-harmony

# Stop both
sudo systemctl stop helixcode-aurora helixcode-harmony

# Status check
sudo systemctl status helixcode-aurora helixcode-harmony

# Enable on boot
sudo systemctl enable helixcode-aurora helixcode-harmony
```

Port Configuration

If running both on the same machine, configure different ports:

Aurora OS (/etc/helixcode/aurora-config.yaml):

```
server:
  port: 8080
```

Harmony OS (/etc/helixcode/harmony-config.yaml):

```
server:
  port: 8081
```

Essential Operations

Creating Your First Task

Aurora OS

```
# Via command line
helixcode-aurora task create \
  --name "Security Scan" \
  --type "analysis" \
  --priority "high"

# Via API
curl -X POST http://localhost:8080/api/tasks \
  -H "Content-Type: application/json" \
  -d '{
```

```
    "name": "Security Scan",
    "type": "analysis",
    "priority": "high"
  }'
```

Harmony OS

Via command line (with distributed computing)

```
helixcode-harmony task create \  
  --name "AI Model Training" \  
  --type "training" \  
  --priority "high" \  
  --distributed \  
  --use-npu
```

Via API

```
curl -X POST http://localhost:8081/api/tasks \  
  -H "Content-Type: application/json" \  
  -d '{  
    "name": "AI Model Training",  
    "type": "training",  
    "priority": "high",  
    "distributed": true,  
    "device": "npu"  
  }'
```

Adding Workers

Aurora OS

Add SSH worker

```
helixcode-aurora worker add \  
  --host 192.168.1.100 \  
  --user helix \  
  --key ~/.ssh/id_rsa
```

List workers

```
helixcode-aurora worker list
```

Harmony OS

Add distributed worker

```
helixcode-harmony worker add \  
  --host 192.168.1.101 \  
  --device npu
```



```
--user helix \  
--key ~/.ssh/id_rsa \  
--capabilities npu,gpu
```

```
# Enable auto-discovery
```

```
helixcode-harmony distributed discovery enable
```

Monitoring

Aurora OS

```
# System status
```

```
helixcode-aurora status
```

```
# Security monitoring
```

```
helixcode-aurora security monitor
```

```
# Resource usage
```

```
helixcode-aurora resources
```

Harmony OS

```
# Cluster status
```

```
helixcode-harmony distributed status
```

```
# Device acceleration
```

```
helixcode-harmony device status
```

```
# Sync status
```

```
helixcode-harmony sync status
```

Platform-Specific Features

Aurora OS Features

Enhanced Security

```
# Check security level
```

```
helixcode-aurora security level
```

```
# Set security level
```

```
helixcode-aurora security level --set enhanced
```

```
# Security audit
helixcode-aurora security audit
```

System Monitoring

```
# Enable advanced monitoring
helixcode-aurora monitor enable --deep-inspection
```

```
# View system metrics
helixcode-aurora monitor metrics
```

```
# Generate report
helixcode-aurora monitor report --format pdf
```

Harmony OS Features

Distributed Computing

```
# Start distributed engine
helixcode-harmony distributed start
```

```
# Add workers to cluster
helixcode-harmony distributed workers add \
  --host worker1.local \
  --host worker2.local
```

```
# Check cluster health
helixcode-harmony distributed health
```

Cross-Device Sync

```
# Enable sync
helixcode-harmony sync enable
```

```
# Configure sync interval
helixcode-harmony sync interval --set 30
```

```
# Force sync
helixcode-harmony sync now --force
```

AI Acceleration

```
# Check NPU availability
helixcode-harmony device npu-status
```

```
# Enable NPU acceleration
helixcode-harmony device npu enable
```

```
# Run accelerated inference
helixcode-harmony infer \
  --model llama-3-8b \
  --device npu \
  --batch-size 4
```

Troubleshooting

Aurora OS Issues

Service Won't Start

```
# Check logs
sudo journalctl -u helixcode-aurora -n 50

# Validate configuration
helixcode-aurora --config /etc/helixcode/aurora-config.yaml --
  validate

# Check database connection
psql -h localhost -U helixcode -d helixcode
```

Security Features Not Working

```
# Verify Aurora OS
cat /etc/aurora-release

# Check security module
helixcode-aurora security diagnose

# Review security logs
tail -f /var/log/helixcode/security.log
```

Harmony OS Issues

Distributed Computing Not Working

```
# Check network
helixcode-harmony network test
```

```
# Verify firewall
sudo firewall-cmd --list-ports
```

```
# Check worker connectivity
helixcode-harmony distributed ping <worker-ip>
```

NPU Not Detected

```
# Check NPU device
ls -la /dev/davinci*
```

```
# Verify Harmony OS
cat /etc/harmonyos-release
```

```
# Update drivers (if needed)
helixcode-harmony device npu-update
```

Common Issues (Both Platforms)

Database Connection Failed

```
# Test database
psql -h localhost -U helixcode -d helixcode
```

```
# Reset password
sudo -u postgres psql -c "ALTER USER helixcode WITH PASSWORD
    'new_password';"
```

```
# Update env file
sudo nano /etc/helixcode/aurora.env # or harmony.env
```

Port Already in Use

```
# Check what's using the port
sudo lsof -i :8080
```

```
# Change port in config
sudo nano /etc/helixcode/aurora-config.yaml # or harmony-
    config.yaml
```

```
# Restart service
sudo systemctl restart helixcode-aurora # or helixcode-harmony
```

Next Steps

Aurora OS

1. **Review Security Configuration:** Read [Aurora OS Security Guide](#)
2. **Configure System Monitoring:** Set up advanced monitoring dashboards
3. **Integrate with Authentication:** Connect to your organization's auth system
4. **Set Up Backups:** Configure automated backup schedules

Harmony OS

1. **Set Up Distributed Cluster:** Add more workers for distributed computing
2. **Enable AI Acceleration:** Configure NPU/GPU for optimal performance
3. **Configure Cross-Device Sync:** Set up Super Device integration
4. **Optimize Resources:** Fine-tune resource management policies

Both Platforms

1. **Add Workers:** Expand your worker pool for better performance
 2. **Configure Workflows:** Set up automated development workflows
 3. **Enable Notifications:** Configure Slack, Discord, or email notifications
 4. **Performance Tuning:** Optimize for your specific workload
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Additional Resources

Documentation

- [Harmony OS Complete Guide](#) - Comprehensive Harmony OS documentation
- [Aurora OS Complete Guide](#) - Comprehensive Aurora OS documentation
- [API Reference](#) - REST API documentation
- [Configuration Reference](#) - All configuration options

Deployment Scripts

- `scripts/deploy-aurora-os.sh` - Aurora OS automated deployment
- `scripts/deploy-harmony-os.sh` - Harmony OS automated deployment
- `scripts/deploy-specialized-platforms.sh` - Combined deployment script

Support

- **Documentation:** <https://docs.helixcode.dev>
- **GitHub Issues:** <https://github.com/helixcode/helixcode/issues>

Quick Reference Card

Aurora OS

Task	Command
Start	<code>sudo systemctl start helixcode-aurora</code>
Stop	<code>sudo systemctl stop helixcode-aurora</code>
Status	<code>sudo systemctl status helixcode-aurora</code>
Logs	<code>sudo journalctl -u helixcode-aurora -f</code>
Config	<code>/etc/helixcode/aurora-config.yaml</code>
Binary	<code>/opt/helixcode/aurora-os</code>
Port	8080 (default)

Harmony OS

Task	Command
Start	<code>sudo systemctl start helixcode-harmony</code>
Stop	<code>sudo systemctl stop helixcode-harmony</code>
Status	<code>sudo systemctl status helixcode-harmony</code>
Logs	<code>sudo journalctl -u helixcode-harmony -f</code>
Config	<code>/etc/helixcode/harmony-config.yaml</code>
Binary	<code>/opt/helixcode/harmony-os</code>
Port	8080 (default)